

DOBERLUG KIRCHHAIN, Germany

WMO: 104900

Lat: 51.6450N

Lon: 13.5747E

Elev: 98

StdP: 100.16

Time Zone: 1.00 (EUC)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.3	-11.2	-18.8	0.7	-15.0	-14.6	1.1	-10.5	13.8	6.0	12.1	4.3	2.0	200	0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	29.7	19.2	27.7	18.2	26.1	17.7	20.2	27.5	19.4	25.9	18.4	24.4	3.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	12.9	23.0	16.8	12.1	21.9	15.9	11.4	21.2	58.5	27.4	55.5	26.0	52.3	24.4	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	9.5	8.4	-15.8	33.9	6.7	2.3	-20.6	35.5	-24.5	36.9	-28.2	38.1	-33.1	39.8
			-16.3	22.7	6.6	0.9	-21.1	23.3	-24.9	23.8	-28.6	24.3	-33.4	24.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	0.0	-0.4	4.4	7.6	12.9	15.6	18.0	17.9	14.2	9.9	4.4	1.8
	DBStd	7.63	5.92	5.43	3.87	3.64	3.51	3.07	2.83	3.27	2.93	3.97	3.89	4.18
	HDD10.0	1358	310	292	174	89	14	1	0	0	2	51	170	254
	HDD18.3	3518	568	524	431	322	171	93	41	48	128	262	418	512
	CDD10.0	966	0	1	2	18	104	168	248	246	128	48	2	1
	CDD18.3	84	0	0	0	0	2	10	31	36	4	1	0	0
	CDH23.3	1122	0	0	0	2	62	150	428	417	61	3	0	0
	CDH26.7	282	0	0	0	0	2	24	117	131	7	0	0	0
Wind	WSAvg	3.9	4.7	4.3	4.4	3.8	3.4	3.5	3.4	3.1	3.5	3.8	4.3	4.4
Precipitation	PrecAvg	602	47	37	42	33	57	59	75	66	50	40	50	47
	PrecMax	856	82	70	114	92	112	121	173	196	145	79	127	90
	PrecMin	386	4	9	11	1	14	8	17	14	6	9	0	21
	PrecStd	112	21	17	27	19	29	28	39	46	28	20	27	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	14.5	17.7	22.7	26.7	29.9	32.4	33.7	27.7	23.1	14.8	12.6
		MCWB	9.0	10.1	11.8	13.7	17.2	18.3	20.3	21.0	17.8	16.7	11.2	9.7
	2%	DB	9.5	11.2	14.4	19.5	25.1	27.3	29.7	30.0	25.1	20.1	12.7	11.3
		MCWB	7.4	8.0	9.6	12.4	16.0	17.7	19.1	19.9	16.7	14.4	10.0	8.5
	5%	DB	8.2	8.3	12.0	17.2	23.3	25.2	27.7	27.6	22.5	17.7	11.2	9.6
		MCWB	6.2	5.8	8.3	11.1	15.0	17.2	18.1	18.5	15.8	13.3	9.1	7.7
	10%	DB	6.9	6.4	10.3	14.7	21.0	23.1	25.9	25.5	20.1	15.8	9.7	7.9
		MCWB	4.9	4.2	7.3	9.7	14.0	16.2	17.8	17.8	14.9	12.1	7.9	6.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	10.3	12.0	14.5	18.8	20.2	22.1	21.7	19.3	17.0	11.7	10.1
		MCDB	10.9	14.2	16.7	21.2	25.1	26.2	29.4	30.8	25.7	22.7	13.8	12.6
	2%	WB	7.7	8.0	10.2	12.9	17.0	19.0	20.4	20.3	17.7	15.1	10.4	8.7
		MCDB	9.4	10.8	13.8	18.9	22.8	25.0	27.7	28.2	23.5	19.1	12.3	10.7
	5%	WB	6.5	6.0	8.8	11.3	15.7	17.8	19.3	19.4	16.4	13.7	9.2	7.8
		MCDB	8.1	8.2	11.5	16.1	21.6	23.6	25.6	26.3	21.3	16.6	10.9	9.6
	10%	WB	5.0	4.4	7.6	10.0	14.5	16.6	18.2	18.3	15.3	12.5	8.0	6.5
		MCDB	6.7	6.4	10.0	14.1	20.2	22.1	24.3	24.2	19.6	15.3	9.7	7.9
Mean Daily Temperature Range	5% DB	MDBR	5.5	7.0	7.9	10.0	11.5	10.2	11.4	11.4	9.7	8.8	6.2	4.8
		MCDBR	5.7	9.3	11.6	14.7	16.3	15.3	16.6	15.9	14.3	12.3	8.2	5.8
		MCWBR	4.7	7.0	7.9	8.7	9.0	7.9	8.1	7.7	8.3	7.7	6.4	4.6
	5% WB	MCDBR	5.8	9.1	11.0	13.0	14.1	12.8	14.2	13.9	12.7	10.7	8.0	5.4
		MCWBR	4.8	6.9	7.8	7.9	8.3	7.2	7.6	7.1	7.5	7.1	6.5	4.6
Clear-Sky Solar Irradiance	taub	0.319	0.344	0.385	0.418	0.419	0.415	0.434	0.427	0.388	0.368	0.343	0.312	
	taud	2.351	2.328	2.227	2.194	2.236	2.277	2.232	2.267	2.348	2.374	2.384	2.365	
	Ebn at Noon	684	769	798	816	835	841	816	800	791	730	647	627	
	Edh at Noon	63	84	112	130	131	127	131	120	99	80	62	54	
All-Sky Solar Radiation	RadAvg	0.72	1.40	2.41	3.98	4.88	5.20	4.91	4.25	3.07	1.77	0.84	0.55	
	RadStd	0.09	0.24	0.30	0.51	0.58	0.42	0.62	0.37	0.38	0.21	0.12	0.06	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.34	+0.38	N/A	N/A	N/A	N/A

Nomenclature: See separate page